

A-level

# ENVIRONMENTAL SCIENCE

## Questions by topic: Aquaculture - Set 1

### Mark scheme

AQA spec 3.5.2.3; Biological resources; textbook pp. 352-359

### Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 01.1     | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• MEDCs: market demand and higher prices for the flavour of carnivorous fish (salmon, trout) justify high inputs (fishmeal, cooler climates suit them) <b>(1)</b></li> <li>• LEDCs: herbivores / omnivores feed on natural vegetation, phytoplankton and wastes so need low inputs; warm climates suit tilapia <b>(1)</b></li> <li>• Herbivores are more energy-efficient (lower trophic level) so more food per unit area <b>(1)</b></li> </ul>                                                                                               | 3     |
| 01.2     | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Some genetically female (XX) fish are treated with testosterone so they become functional males producing only X-carrying sperm <b>(1)</b></li> <li>• Crossed with normal females (XX) all offspring are XX females <b>(1)</b></li> <li>• Female trout meat has a better flavour; in tilapia the reverse (all males) because males grow larger / more efficiently <b>(1)</b></li> </ul>                                                                                                                                                      | 3     |
| 01.3     | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Salmon need high oxygen: water sprays / weirs aerate; wastes removed to prevent deoxygenation <b>(1)</b></li> <li>• Constant circular flow keeps fish swimming the same way - fewer collisions / injuries, more muscular fish, higher stocking density <b>(1)</b></li> <li>• Artificial lighting delays maturation (salmon stop growing at maturity) so larger fish; controls smoltification <b>(1)</b></li> <li>• Cool sites (Scotland, Norway) because warm water holds less oxygen although warmth increases growth <b>(1)</b></li> </ul> | 4     |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 352-355; AQA spec 3.5.2.3.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marks |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>02.1</b> | <p><i>Up to 4 marks in total; award marks in proportion to the completeness of each point.</i></p> <ul style="list-style-type: none"> <li>• Extensive: low inputs of energy, food and chemicals so low cost and low environmental impact; sustainable; but low yield per area and dependent on natural productivity <b>(up to 2)</b></li> <li>• Intensive: control of all limiting factors gives very high yield and profit per area; but high inputs (fishmeal from wild fish, energy), disease and lice at high density, pollution from wastes, antibiotic / pesticide use, escapes <b>(up to 2)</b></li> <li>• Judgement depends on species, market and site <b>(up to 2)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4     |
| <b>02.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Adults with desirable characteristics (fast growth, shape, colour) stripped of eggs and milt which are mixed for fertilisation <b>(1)</b></li> <li>• Eggs raised in aerated freshwater tanks, dead / diseased eggs removed daily; fry fed fishmeal pellets and moved through tanks as they grow <b>(1)</b></li> <li>• After 12-18 months smolts transferred to seawater cages, fed pellets until harvest at 3-5 kg <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3     |
| <b>02.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• About 10% of the world's mangroves have been cleared for coastal shrimp lagoons (up to 100 ha), removing nurseries and coastal protection <b>(1)</b></li> <li>• Lagoons need water exchange to remove wastes and aeration to prevent deoxygenation as wastes decompose <b>(1)</b></li> <li>• Eyestalk ablation stimulates breeding; high densities increase disease so antibiotics / pesticides used <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 3     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-356; AQA spec 3.5.2.3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |       |
| <b>03.1</b> | <p><i>2 marks for each point (partial credit for a partly correct point).</i></p> <ul style="list-style-type: none"> <li>• Carnivores fed pellets of sandeels, herring, anchovy; overfishing sandeels reduces puffins and terns; reduced by more plant protein / vegetable oil in pellets or farming herbivores (tilapia) <b>(2)</b></li> <li>• High density spreads lice (which spread to wild fish) and infections; pesticides (organophosphates, pyrethroids, hydrogen peroxide) kill wildlife; antibiotics breed resistant bacteria; reduced by lower densities, single-age tanks with flow from young to old, wrasse as biological control, warm-water washes, circulating flow <b>(2)</b></li> <li>• Escaped farmed fish breed with wild fish introducing disadvantageous genes (bright colour), or non-indigenous escapees colonise as predators / competitors; reduced by better cage design, sterile triploid fish (pressure-treated eggs), species that cannot breed locally (Japanese oysters in UK) <b>(2)</b></li> <li>• Faeces and uneaten food decompose causing deoxygenation under cages; reduced by monitored feeding, siting cages in strong currents, effluent treatment / reed beds for freshwater farms, IMTA with seaweed and shellfish absorbing wastes <b>(2)</b></li> </ul> | 8     |
| <b>03.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Predators are killed under licence to protect stock, reducing wild populations of protected / valued species <b>(1)</b></li> <li>• Better cage designs, netting, fencing, bird scarers exclude predators without killing <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 356-359; AQA spec 3.5.2.3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |       |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks |
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| <b>04.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Silver carp eat phytoplankton, bighead carp zooplankton, grass carp and tilapia vegetation, common carp bottom-feed on molluscs / insects, catfish or perch eat small fish <b>(1)</b></li> <li>• Species occupy different niches / trophic levels so they do not compete and all food sources are used <b>(1)</b></li> <li>• Bottom feeders re-suspend nutrients boosting plankton / plant growth <b>(1)</b></li> <li>• Predators control over-breeding species (tilapia) so remaining fish grow larger; total yield exceeds any single species <b>(1)</b></li> </ul>                                                                                                          | 4     |
| <b>04.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• IMTA: fed species (salmon, shrimp) alongside inorganic extractive species (seaweeds absorbing ammonia and phosphate) and organic extractive species (filter-feeding mussels eating waste particles / plankton) <b>(1)</b></li> <li>• Wastes of one species become nutrients for another so less pollution and extra harvests <b>(1)</b></li> <li>• Aquaponics: aquaculture water rich in nutrients / organic matter feeds hydroponic salad crops (lettuce, herbs, watercress) <b>(1)</b></li> <li>• Plants remove the nutrients that would cause deoxygenation and cleaned water returns to the fish (tilapia, carp, catfish); rice-fish paddies similar <b>(1)</b></li> </ul> | 4     |
| <b>04.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Bluefin tuna do not breed well in captivity so young fish are caught from the wild and fattened in sea cages <b>(1)</b></li> <li>• Production still depends on (and removes juveniles from) the wild population; carnivores need large fish inputs <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                       | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 357-358; AQA spec 3.5.2.3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks |
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| 05.1     | <p><b>Levels of response</b></p> <p><b>Level 3 (7-9 marks):</b> Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.</p> <p><b>Level 2 (4-6 marks):</b> Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.</p> <p><b>Level 1 (1-3 marks):</b> Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (credit other relevant, accurate points)</p> <ul style="list-style-type: none"> <li>• Context: wild catches have not increased since about 1990 as stocks are fully or over-exploited; aquaculture output growing rapidly and approaching the wild catch; fish are the last large-scale hunted food</li> <li>• Advantages of aquaculture: control of limiting factors, selective breeding, gender control, high yields per area; fish have low FCRs (no thermoregulation / support) compared with livestock; herbivores (tilapia, carp) and molluscs very efficient; polyculture, IMTA, aquaponics, rice-fish</li> <li>• Limits: carnivorous species (salmon, shrimp) need fishmeal from wild fish so still depend on fishing and deplete forage fish (sandeels - puffins); some species (bluefin tuna) will not breed in captivity; consumer preference for carnivores</li> <li>• Environmental impacts: habitat loss (mangroves), organic pollution, pesticides, antibiotics and resistance, lice, escapes and gene pool contamination, predator control; and how they can be reduced</li> <li>• Fishing can be sustainable with quotas, gear design, NTZs, size limits, effort limits; MSY management needs good data</li> <li>• Judgement: aquaculture can supply much of the growth in demand, especially herbivores and shellfish, but cannot fully replace fishing; both need sustainable management</li> </ul> | 9     |
| 05.2     | <p><i>1 mark.</i></p> <ul style="list-style-type: none"> <li>• A species that is given artificial food (pellets) rather than relying on natural food in the water, eg salmon, shrimp <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1     |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 338-359; AQA spec 3.5.2.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |       |